Here θ is the angle between $\vec{B}$ and the length vector $\vec{l}$ for the left segment and is 90°

$$\therefore F_1 = ilB\sin 90^\circ = ilB \qquad \text{since } \sin 90^\circ = 1$$

The applied force $\vec{F}$ must be equal to $\vec{F}_1$ in order to just move the loop with a constant velocity $\vec{v}$

$$\therefore \vec{F} = -\vec{F}_1$$

$\left(\text{since } \vec{F} \text{ and } \vec{F}_1 \text{ are in opposite direction}\right)$

Considering only the magnitudes,

$$F = F_1 = ilB$$

Substituting for i from equation (4.10)

$$F = \left(\frac{Blv}{R}\right)lB$$

$$F = \frac{B^2l^2v}{R}$$

From equation (4.11), the rate at which the mechanical work is done to pull the loop from the magnetic field or power is given by

$$P = Fv = \left(\frac{B^2l^2v}{R}\right)v$$

$$P = \frac{B^2l^2v^2}{R} \qquad (4.13)$$

When the induced current flows in the loop, Joule heating takes place. The rate at which thermal energy is dissipated in the loop or power dissipated is

$$P = i^2R$$

$$P = \left(\frac{Blv}{R}\right)^2 R$$

$$P = \frac{B^2l^2v^2}{R} \qquad (4.14)$$

This equation is exactly same as the equation (4.13). **Thus the mechanical work done in moving the loop appears as thermal energy in the loop**.

EXAMPLE 4.8

A conducting rod of length 0.5 m falls freely from the top of a building of height 7.2 m at a place in Chennai where the horizontal component of Earth's magnetic field is 40378.7 *nT*. If the length of the rod is perpendicular to Earth's horizontal magnetic field, find the emf induced across the conductor when the rod is about to touch the ground. [Take g = 10 m s^{-2}]

Solution

l = 0.5 m; h = 7.2 m; u = 0 m s^{-1};
g = 10 m s^{-2}; B_H = 40378.7 *nT*

The final velocity of the rod is

$$v^2 = u^2 + 2gh$$

$$= 0 + (2\times 10\times 7.2)$$

$$v^2 = 144$$

$$v = 12\,ms^{-1}$$

Induced emf when the rod is about to touch the ground, $\varepsilon = B_H lv$

$$= 40{,}378.7\times 10^{-9}\times 0.5\times 12$$

$$= 242.27\times 10^{-6}V$$

$$= 242.27\mu V$$

EXAMPLE 4.9

A copper rod of length l rotates about one of its ends with an angular velocity ω in a magnetic field B as shown in the figure. The plane of rotation is perpendicular to the field. Find the emf induced between the two ends of the rod.

UNIT-4(XII-Physics_Vol-1).indd 220 04-03-2019 10:39:15